

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0674

Trans Mountain 138 kV
February 17, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0674. The Interconnection Customer proposes a 200.1 MW wind facility interconnecting at the Trans Mountain 138 kV substation in the Virginia area. The request was received on September 09, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 282 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$48,081,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CARMEL 146TH ST / OAK CRK 138 kV	BEAR / CLOVER 345 kV line	143.1	200	6.5
CEIBA BAJA 7012 / CARMEL 146TH T 138 kV	BEAR / CLOVER 345 kV Line	142.0	204	6.5
TANGY / CEIBA BAJA 7012 138 kV Line	BEAR / CLOVER 345 kV line	131.0	229	6.5
ULTRAGEN / TRANS MTN 138 kV	MONEL / TRANS MTN 138 kV	122.7	155	100.0
NEW LONDON / BARKSDALE 138 kV	WYTHE / S CREEK SUBSTATION 138 kV	119.6	150	9.2
CARSON ICE COGEN SUBSTATION / ALDERWOOD 138 kV	REACH / CANDELARIA 138 kV L	118.7	233	11.9
NEW LONDON / FRANKLIN-ELK GROVE 138 kV Line	REACH / CANDELARIA 138 kV Line	117.4	357	31.5
GRASSMERE / RED HILL 138 kV	BEAR / CLOVER 345 kV	114.8	227	7.4
FRANKLIN-ELKGROVE / BANNING 138 kV	FRANKLIN-ELK GROVE / SOUTH TEXA 138 kV	114.5	215	23.1
COAL MTN SUBSTATION / GRASSMERE 138 kV	BEAR / CLOVER 345 kV line	112.9	257	7.4
COAL MTN / CENTRAL AUSTIN 138 kV	BEAR / CLOVER 345 kV	112.1	266	7.4
CENTRAL AUSTIN / WAYNESBORO 138 kV	BEAR / CLOVER 345 kV	111.2	277	7.4
BANNING / DESERT OUTPOST 138 kV Line	S TEXA / FRANKLIN-ELK GROVE 138 kV	109.9	234	23.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ALDERWOOD / WAYNESBORO SUBSTATION 138 kV	CLOVER / BEAR 345 kV	108.9	305	7.4
REACH / CANDELARIA 138 kV	BEAR / CLOVER 345 kV Line	105.0	468	16.9
ASPEN / FAIRFIELD 138 kV	ASPEN / LOWMOOR 345 kV Line	104.6	214	7.3
TRANS MTN / MONEL 138 kV L	ULTRAGEN / TRANS MTN 138 kV line	102.8	185	100.0
BEAR SUBSTATION / LEAVITT 345 kV	BEAR / LEAVITT 345 kV #1	101.4	1,957	13.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CARMEL 146TH ST / OAK CRK 138 kV	Rebuild line with higher-capacity conductor	\$6,505,000	15% (\$999,000)
CEIBA BAJA 7012 / CARMEL 146TH T 138 kV	Rebuild line with higher-capacity conductor	\$10,870,000	15% (\$1,669,000)
TANGY / CEIBA BAJA 7012 138 kV Line	Rebuild line with higher-capacity conductor	\$10,594,000	15% (\$1,627,000)
ULTRAGEN / TRANS MTN 138 kV	Rebuild line with higher-capacity conductor	\$22,197,000	100% (\$22,197,000)
NEW LONDON / BARKSDALE 138 kV	Rebuild line with higher-capacity conductor	\$11,046,000	22% (\$2,475,000)
ALDERWOOD / CARSON ICE COGEN 138 kV	Rebuild line with higher-capacity conductor	\$4,770,000	17% (\$800,000)
NEW LONDON / FRANKLIN-ELK GROVE 138 kV Line	Rebuild line with higher-capacity conductor	\$2,894,000	14% (\$411,000)
GRASSMERE / RED HILL 138 kV	Rebuild line with higher-capacity conductor	\$2,603,000	10% (\$256,000)
FRANKLIN-ELK GROVE / BANNING 138 kV	Rebuild line with higher-capacity conductor	\$2,678,000	16% (\$437,000)
COAL MTN / GRASSMERE 138 kV line	Rebuild line with higher-capacity conductor	\$3,728,000	10% (\$367,000)
COAL MTN / CENTRAL AUSTIN 138 kV	Rebuild line with higher-capacity conductor	\$6,060,000	10% (\$596,000)
CENTRAL AUSTIN / WAYNESBORO 138 kV	Rebuild line with higher-capacity conductor	\$3,687,000	10% (\$363,000)
BANNING / DESERT OUTPOST 138 kV Line	Reconductor line	\$2,328,000	16% (\$380,000)
ALDERWOOD / WAYNESBORO SUBSTATION 138 kV	Reconductor line	\$2,131,000	10% (\$210,000)
REACH / CANDELARIA 138 kV	Reconductor line	\$5,714,000	9% (\$514,000)
ASPEN / FAIRFIELD 138 kV	Reconductor line	\$5,514,000	30% (\$1,662,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MONEL / TRANS MN 138 kV	Reconductor line	\$10,478,000	100% (\$10,478,000)
LEAVITT SUBSTATION / BEAR 345 kV #2	Reconductor line	\$78,647,000	3% (\$2,640,000)
	Total allocated to project		\$48,081,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.